API Documentation

Base URL: http://127.0.0.1:8000. Interactive docs: /docs (Swagger UI) and /openapi.json. All routes are generated from the running application; validation errors return HTTP 422 with a list of {loc, msg}; domain errors return {"detail", "code"}.

Status codes

Code	Meaning
200 / 201 / 202	Success / created (feedback) / accepted background job
400	Invalid request value (e.g. unknown sort_by)
404	Unknown case, query, run or job
409	An evaluation is already running
413	Payload larger than 64 KB
422	Schema validation failure (ranges, lengths, enums)
503	Dataset, index or model artifact missing / retrieval backends unavailable (code field says which)

Endpoint reference

Method	Path	Purpose	Parameters / body
POST	/api/advisory/classify	Advisory Classify	**ClassifyRequest**: query, use_llm
POST	/api/advisory/jobs	Advisory Job Start - Start the advisory workflow in the background; poll GET /advisory/jobs/{job_id} for real stage progress.	**AdvisoryRequest**: query, crop, state, season, n, p, k, ph, rainfall, temperature, humidity, fertilizer, pesticide, area, top_k, max_recommendations, use_llm, include_synthetic
GET	/api/advisory/jobs/{job_id}	Advisory Job Status	job_id
GET	/api/advisory/queries/{query_id}	Advisory Get	query_id
POST	/api/advisory/query	Advisory Query - Run the full multi-agent advisory workflow synchronously.	**AdvisoryRequest**: query, crop, state, season, n, p, k, ph, rainfall, temperature, humidity, fertilizer, pesticide, area, top_k, max_recommendations, use_llm, include_synthetic
GET	/api/advisory/recent	Advisory Recent	limit
POST	/api/advisory/recommend	Advisory Recommend - Same pipeline, returning only the recommendation-focused part of the response.	**AdvisoryRequest**: query, crop, state, season, n, p, k, ph, rainfall, temperature, humidity, fertilizer, pesticide, area, top_k, max_recommendations, use_llm, include_synthetic
POST	/api/analysis/impact	Analysis Impact	**ImpactRequest**: crop, state, season
GET	/api/cases	List Cases - Filter via query params: crop, state, season (repeatable), year_min/max, and _min/_max for	q, sort_by, order, page, page_size
POST	/api/cases/search	Search Cases - Hybrid (BM25 + semantic, RRF) search with optional metadata filters.	**SearchRequest**: query, filters, top_k, mode
GET	/api/cases/{case_id}	Get Case	case_id
GET	/api/evaluations	Evaluations Overview - Latest evaluation summary. Returns status 'not_evaluated' until a run exists.	
POST	/api/evaluations/run	Evaluation Run Start	label, deepeval_limit, use_llm
GET	/api/evaluations/runs	Evaluation Runs	
GET	/api/evaluations/runs/{run_id}	Evaluation Run	run_id
GET	/api/evaluations/test-cases	Evaluation Test Cases	
POST	/api/feedback	Post Feedback	**FeedbackIn**: query_id, recommendation_id, crop, helpful, rating, category, comment, case_ids
GET	/api/feedback	Get Feedback	page, page_size
GET	/api/feedback/summary	Feedback Summary	

Method	Path	Purpose	Parameters / body
GET	/api/health	Api Health	
GET	/api/insights/advisory-summary	Advisory Summary - Aggregates over persisted advisory runs: judge verdicts, risk categories, agent outcomes.	limit
GET	/api/metadata/options	Metadata Options	
POST	/api/predictions/suitability	Predict Suitability	**SuitabilityRequest**: crop, state, season, n, p, k, ph, rainfall, temperature, humidity
POST	/api/predictions/yield	Predict Yield	**YieldRequest**: crop, state, season, year, n, p, k, ph, rainfall, temperature, humidity, area, fertilizer, pesticide
GET	/api/stats	Stats	
GET	/api/system/status	System Status	
GET	/health	Health	

Sample requests and real responses

Captured from the running application with use_llm=false (so no API credit is used). Long lists are truncated with '...'.

POST /api/advisory/query - HTTP 200

```
Request: {"query": "Which crops perform well in Punjab during Rabi with soil pH 7.8 and rainfall around 600 mm?", "use_
{
 "query_id": "q-f912fbbef3",
 "status": "completed_with_warnings",
 "query": "Which crops perform well in Punjab during Rabi with soil pH 7.8 and rainfall around 600 mm?",
 "validated_input": {
  "query": "Which crops perform well in Punjab during Rabi with soil pH 7.8 and rainfall around 600 mm?",
  "top_k": 2,
  "max_recommendations": 2,
  "use_llm": false,
  "include_synthetic": true
 },
 "conditions": {
  "crop": {
   "value": null,
   "provenance": "missing",
   "note": null
  },
  "state": {
   "value": "Punjab",
   "provenance": "parsed_from_query",
   "note": null
  },
  "season": {
   "value": "Rabi",
   "provenance": "parsed_from_query",
   "note": null
  },
  "n": {
   "value": null,
   "provenance": "missing",
   "note": null
  },
  "p": {
   "value": null,
   "provenance": "missing",
   "note": null
  },
  "k": {
   "value": null,
   "provenance": "missing",
   "note": null
  },
  "ph": {
   "value": 7.8,
   "provenance": "parsed_from_query",
   "note": null
  },
  "rainfall": {
   "value": 600.0,
   "provenance": "parsed_from_query",
   "note": null
  },
  "temperature": {
   "value": null,
   "provenance": "missing",
   "note": null
```

```
  },
  "humidity": {
   "value": null,
   "provenance": "missing",
   "note": null
  },
  "fertilizer": {
   "value": null,
   "provenance": "missing",
   "note": null
  },
  "pesticide": {
   "value": null,
   "provenance": "missing",
   "note": null
  },
  "excluded_crops": [],
  "objectives": []
 },
 "retrieval": {
  "mode": "hybrid",
  "filters_applied": {
   "season": [
    "Rabi"
   ],
   "state": [
    "Punjab"
   ]
  },
  "backends": {
   "keyword": true,
   "semantic": true,
   "errors": {},
   "used": {
    "keyword": true,
    "semantic": true
   }
  },
  "fallback": null,
  "total_candidates_after_filter": 393,
  "filtered_out": 54607,
  "scoring": "Hybrid score = Reciprocal Rank Fusion: sum(1/(60+rank)) over BM25 and semantic rankings. It is a rank-bas
  "filter_diagnostics": {},
  "priority_formula": "see src/rag/prioritizer.py"
 },
 "cases": [
  {
   "case_id": "TM-CCCF09D177",
   "hybrid_rank": 1,
   "hybrid_score": 0.02765,
   "keyword_rank": 9,
   "keyword_score": 49.0116,
   "semantic_rank": 16,
   "semantic_score": 0.8224,
   "matched_terms": [
    "alkaline",
    "low",
    "..."
   ],
   "relevance_explanation": "keyword rank 9 (BM25 49.01; matched: alkaline, low, mm, ph, punjab, rabi, rainfall, soil)
   "record": {
    "case_id": "TM-CCCF09D177",
    "source_row": 9941,
    "crop": "Gram",
    "year": 2006,
    "
```


POST /api/cases/search - HTTP 200

Request: {"query": "wheat punjab rabi", "top_k": 2, "filters": {"state": ["Punjab"]}}


```
{
 "query": "wheat punjab rabi",
 "mode": "hybrid",
 "items": [
  {
   "case_id": "TM-EB5AB3175F",
   "hybrid_rank": 1,
   "hybrid_score": 0.03102,
   "keyword_rank": 3,
   "keyword_score": 8.3051,
   "semantic_rank": 6,
   "semantic_score": 0.8226,
   "matched_terms": [
    "punjab",
    "rabi",
```

```
      "..."
     ],
     "relevance_explanation": "keyword rank 3 (BM25 8.31; matched: punjab, rabi, wheat); semantic rank 6 (cosine 0.823)",
     "record": {
      "case_id": "TM-EB5AB3175F",
      "source_row": 12354,
      "crop": "Wheat",
      "year": 2011,
      "season": "Rabi",
      "state": "Punjab",
      "area": 3528000.0,
      "production": 17982000.0,
      "yield": 5.0675,
      "fertilizer": 591010560.0,
      "pesticide": 1164240.0,
      "n": 150,
      "p": 50,
      "k": 40
     },
     "priority_score": null,
     "priority_components": null,
     "priority_explanation": null
    },
    {
     "case_id": "TS-8FD5F1E637",
     "hybrid_rank": 2,
     "hybrid_score": 0.02879,
     "keyword_rank": 19,
     "keyword_score": 8.3051,
     "semantic_rank": 2,
     "semantic_score": 0.8238,
     "matched_terms": [
      "punjab",
      "rabi",
      "..."
     ],
     "relevance_explanation": "keyword rank 19 (BM25 8.31; matched: punjab, rabi, wheat); semantic rank 2 (cosine 0.824)",
     "record": {
      "case_id": "TS-8FD5F1E637",
      "source_row": -11141,
      "crop": "Wheat",
      "year": 2001,
      "season": "Rabi",
      "state": "Punjab",
      "area": 3761543.8,
      "production": 16452045.0,
      "yield": 4.373748,
      "fertilizer": 367939349.77,
      "pesticide": 970424.0,
      "n": 150,
      "p": 50,
      "k": 40
     },
     "priority_score": null,
     "priority_components": null,
     "priority_explanation": null
    }
   ],
   "total_candidates_after_filter": 1105,
   "filtered_out": 53895,
   "filters_applied": {
    "state": [
     "Punjab"
    ]
   },
   "backends": {
    "keyword": true,
    "semantic": true,
    "errors": {},
    "used": {
     "keyword": true,
     "semantic": true
    }
   },
   "fallback": null,
   "scoring": "Hybrid score = Reciprocal Rank Fusion: sum(1/(60+rank)) over BM25 and semantic rankings. It is a rank-base
  }
```


POST /api/predictions/yield - HTTP 200


```
Request: {"crop": "Wheat", "state": "Punjab", "season": "Rabi", "ph": 8, "rainfall": 600}
{
 "predicted_yield": 4.3575,
 "unit": "same unit as the source 'Yield' column for the given crop (typically tonnes/hectare; crop-specific)",
 "is_estimate": true,
```

```
 "interval_10_90": {
  "low": 2.1408,
  "high": 5.1509,
  "method": "quantile gradient boosting (log scale), empirical coverage on held-out test years: 73% (nominal 80%)"
 },
 "feature_set": "core",
 "model_name": "hgb",
 "model_version": "yield-20261009155757",
 "assumptions": [
  "year assumed 2020",
  "n not supplied (left missing)",
  "..."
 ],
 "unseen_categories": [],
 "limitations": [
  "Soil N/P/K/pH are state constants, so they act as state identifiers; importance is confounded with state.",
  "Weather is a state-year aggregate; no farm-level data (soil type, irrigation, variety) is available.",
  "..."
 ],
 "test_metrics": {
  "mae": 17.041921798341008,
  "rmse": 290.32132585405793,
  "r2": 0.8830437819752356,
  "mae_log1p": 0.14104410242880755,
  "rmse_log1p": 0.23139811732963483,
  "r2_log1p": 0.956514866120731
 }
}
```


POST /api/predictions/suitability - HTTP 200


```
Request: {"crop": "Wheat", "state": "Punjab", "season": "Rabi", "ph": 8}
{
 "crop": "Wheat",
 "method": "rule_based_historical_envelope",
 "is_trained_classifier": false,
 "description": "Heuristic comparison of supplied conditions with the range in which this crop was historically record
 "category": "low",
 "score": 0.875,
 "factors": [
  {
   "factor": "Soil pH",
   "field": "ph",
   "user_value": 8.0,
   "historical_p10": 5.6,
   "historical_median": 6.5,
   "historical_p90": 7.8,
   "historical_min": 5.5,
   "historical_max": 8.0,
   "score": 0.5,
   "status": "inside recorded range but at the edge (outside P10-P90)"
  },
  {
   "factor": "Season",
   "field": "season",
   "user_value": "Rabi",
   "score": 1.0,
   "status": "94% of this crop's records are in this season"
  },
  "..."
 ],
 "supporting_factors": [
  "Season",
  "Region (state)",
  "..."
 ],
 "risk_factors": [
  "Soil pH: inside recorded range but at the edge (outside P10-P90)"
 ],
 "missing_conditions": [
  "Nitrogen (N)",
  "Phosphorus (P)",
  "..."
 ],
 "limitations": [
  "State-level soil and state-year weather are coarse proxies for field conditions.",
  "Historical presence does not imply agronomic suitability for a specific farm.",
  "..."
 ]
}
```


POST /api/advisory/classify - HTTP 200


```
Request: {"query": "wheat in Punjab Rabi pH 7.8", "use_llm": false}
{
```

```
 "conditions": {
  "crop": {
   "value": "Wheat",
   "provenance": "parsed_from_query",
   "note": null
  },
  "state": {
   "value": "Punjab",
   "provenance": "parsed_from_query",
   "note": null
  },
  "season": {
   "value": "Rabi",
   "provenance": "parsed_from_query",
   "note": null
  },
  "n": {
   "value": null,
   "provenance": "missing",
   "note": null
  },
  "p": {
   "value": null,
   "provenance": "missing",
   "note": null
  },
  "k": {
   "value": null,
   "provenance": "missing",
   "note": null
  },
  "ph": {
   "value": 7.8,
   "provenance": "parsed_from_query",
   "note": null
  },
  "rainfall": {
   "value": null,
   "provenance": "missing",
   "note": null
  },
  "temperature": {
   "value": null,
   "provenance": "missing",
   "note": null
  },
  "humidity": {
   "value": null,
   "provenance": "missing",
   "note": null
  },
  "fertilizer": {
   "value": null,
   "provenance": "missing",
   "note": null
  },
  "pesticide": {
   "value": null,
   "provenance": "missing",
   "note": null
  },
  "excluded_crops": [],
  "objectives": []
 },
 "notes": {
  "llm": "not_used"
 }
}
```


Validation error example - POST /api/advisory/query {"query": "", "ph": 99}


```
HTTP 422
{
 "detail": [
  {
   "type": "string_too_short",
   "loc": [
    "body",
    "query"
   ],
   "msg": "String should have at least 3 characters",
   "input": "",
   "ctx": {
    "min_length": 3
   }
```

```
    },
    {
     "type": "less_than_equal",
     "loc": [
      "body",
      "ph"
     ],
     "msg": "Input should be less than or equal to 10",
     "input": 99,
     "ctx": {
      "le": 10.0
     }
    }
   ]
}
```

Advisory response structure

Field	Content
query_id, status	completed | completed_with_warnings | needs_clarification | no_evidence | failed
validated_input	request after validation
conditions	per-field {value, provenance, note}, missing_information, excluded_crops, objectives
retrieval	filters applied, candidates after filter / filtered out, backends, fallback, scoring note
cases[]	case_id, record, keyword/semantic rank+score, hybrid rank/score, priority_score + components + explanation
recommendations[]	rank, crop, score (heuristic), reasons, supporting_case_ids, historical_yield (actual), yield_prediction (model estimate + interval + assumptions), risk (rule-based), judge, warnings
narrative	LLM or template explanation with source
impact_analysis	correlations, group comparisons, permutation importance, confounding notes
judge	deterministic checks + LLM verdict (evidence_support, condition_match, constraint_violations, unsupported_claims, ...)
agents[], stages[]	real status and duration of each agent / workflow stage
warnings, metadata	limitations, durations, dataset version